

Warm-Up: One Problem, Three Ways

Let $\vec{F} = \langle 2xy + 1, x^2 \rangle$ and let C be the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.

Evaluate $\int_C \vec{F} \cdot d\vec{r}$ using all three methods:

1. **Definition:** Parameterize C , compute $\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t)$, integrate.
2. **Component form:** Write as $\int_C P dx + Q dy$ and substitute.
3. **FTC:** Find a potential function f with $\nabla f = \vec{F}$, then use $f(\text{end}) - f(\text{start})$.

Try all three. You should get the same answer each time. Which method is easiest?

Line Integrals Meet Double Integrals

In 16.3, conservative fields gave us $\oint_C \vec{F} \cdot d\vec{r} = 0$ for every closed curve. But what if $\vec{F}$ is **not** conservative?

Green's Theorem answers this: the line integral around a closed curve equals a **double integral** over the enclosed region.

$$(\text{integral around boundary}) = (\text{integral over interior})$$

This is a higher-dimensional FTC: just as $\int_a^b f'(x) dx = f(b) - f(a)$ relates an interval integral to boundary values, Green's Theorem relates a region integral to a boundary integral.

Positive Orientation

Positive Orientation

A simple closed curve C is **positively oriented** if, as you traverse it, the enclosed region D stays on your **left**. In the plane: **counterclockwise**.

Positive orientation (CCW) Negative orientation (CW)



Clockwise traversal is negative orientation: $\oint_{-C} \vec{F} \cdot d\vec{r} = - \oint_C \vec{F} \cdot d\vec{r}$.

Green's Theorem requires **positive (CCW) orientation**. If the curve is CW, negate.

Green's Theorem

Theorem: Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve bounding a region D . If P and Q have continuous partial derivatives on an open region containing D , then

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

In words: circulation around boundary = total curl over interior.

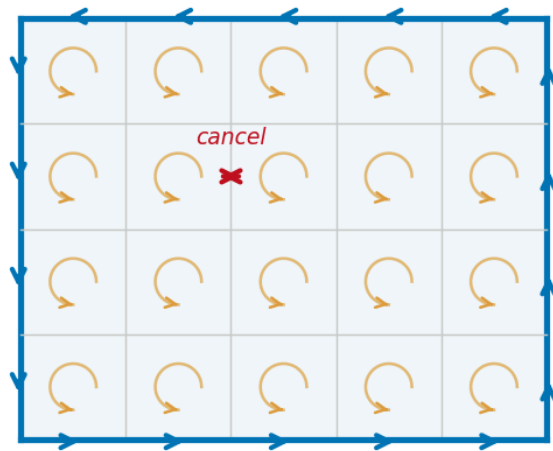
When the curl is constant (say $Q_x - P_y = k$), the integral is just $k \cdot \text{Area}(D)$.

Quick Check: For $\oint_C (3y) dx + (5x) dy$: $P = 3y$, $Q = 5x$, so $Q_x - P_y = 5 - 3 = 2$. By Green's:
 $\oint = \iint_D 2 dA = 2 \cdot \text{Area}(D)$. When the curl is constant, the double integral is just the constant times the area!

Green's Theorem converts a line integral (hard) into a double integral (often easier), or vice versa. Next: **why** does it work?

Why Green's Theorem Works: Cancellation

To see why, break D into tiny pieces.



Tile D with tiny rectangles. Traverse each cell CCW and call its line integral the **micro-circulation** $\Delta\Gamma$.

Interior edges cancel. Every shared edge is traversed in opposite directions by adjacent cells $\rightarrow$ the two contributions sum to zero.

Only the outer boundary C survives:

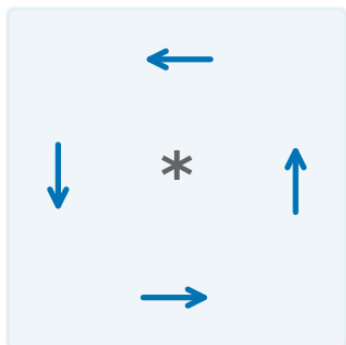
$$\oint_C \vec{F} \cdot d\vec{r} = \sum_{\text{cells}} \Delta\Gamma$$

The boundary circulation equals the **sum of the micro-circulations** inside. Next question: what is $\Delta\Gamma$ on a single tiny cell?

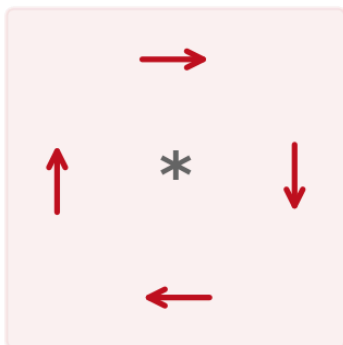
What Does One Cell's Circulation Look Like?

Imagine placing a tiny paddle wheel at a point in a vector field. Does it spin?

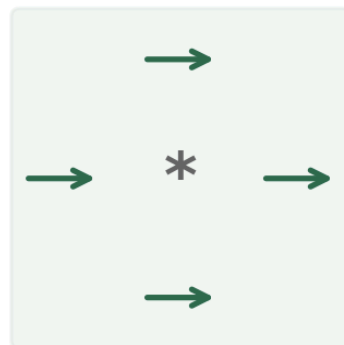
Positive curl
(CCW rotation)



Negative curl
(CW rotation)



Zero curl
(no rotation)



Look at how the arrows change across the box:

- Q **increases** left-to-right ($Q_x > 0$): the right side pushes up more than the left $\rightarrow$ CCW spin
- P **increases** bottom-to-top ($P_y > 0$): the top pushes right more than the bottom $\rightarrow$ CW spin

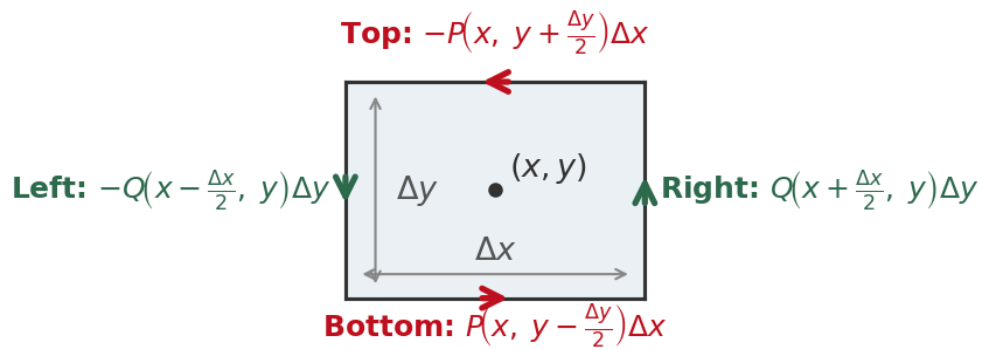
The net rotation is the competition between these: $Q_x - P_y$.

- $Q_x - P_y > 0$: net CCW rotation (positive curl)
- $Q_x - P_y < 0$: net CW rotation (negative curl)
- $Q_x - P_y = 0$: no net rotation (conservative!)

The **2D curl** $Q_x - P_y$ measures local counterclockwise rotation. It's the same quantity from the cross-partials test: zero curl = conservative.

Making It Precise: Circulation on a Tiny Rectangle

Center a tiny rectangle at (x, y) and compute $\oint \vec{F} \cdot d\vec{r}$ around it (CCW):



Horizontal edges (top and bottom) contribute:

$$P\left(x, y - \frac{\Delta y}{2}\right) \Delta x - P\left(x, y + \frac{\Delta y}{2}\right) \Delta x \approx -\frac{\partial P}{\partial y} \Delta y \Delta x$$

Vertical edges (right and left) contribute:

$$Q\left(x + \frac{\Delta x}{2}, y\right) \Delta y - Q\left(x - \frac{\Delta x}{2}, y\right) \Delta y \approx \frac{\partial Q}{\partial x} \Delta x \Delta y$$

One cell's micro-circulation:

$$\Delta \Gamma \approx \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \Delta A$$

Plug into cancellation and take $\Delta A \rightarrow 0$:

$$\oint_C \vec{F} \cdot d\vec{r} = \sum_{\text{cells}} \Delta \Gamma \approx \sum_{\text{cells}} (Q_x - P_y) \Delta A \longrightarrow \iint_D (Q_x - P_y) dA$$

That's Green's Theorem. ✓

Each tiny cell circulates by $(Q_x - P_y) \Delta A$. Sum over all cells: interiors cancel, the boundary survives, and the sum becomes the double integral of the 2D curl.

Example 1: Verify Green's Theorem

Example 1. Evaluate $\oint_C xy \, dx + x^2 \, dy$ where C is the rectangle $(0, 0) \rightarrow (3, 0) \rightarrow (3, 1) \rightarrow (0, 1)$ (CCW). Verify both sides.

Compute $Q_x - P_y$ and integrate over the rectangle. Then verify by computing the line integral along all four edges.

Example 2: Symmetry Shortcut

Example 2

Use Green's Theorem to evaluate $\oint_C y^4 \, dx + 2xy^3 \, dy$ where C is the ellipse $x^2 + 2y^2 = 2$ (CCW).

Compute $Q_x - P_y$. Before integrating, check: is the integrand odd or even on a symmetric region?

Example 3: Green's Theorem Throws Away the Hard Parts

Example 3

Evaluate $\oint_C (e^x + y^2) dx + (e^y + x^2) dy$ where C is the boundary of the region between $y = x^2$ and $y = x$ (CCW).

Compute $Q_x - P_y$. Find where $y = x^2$ and $y = x$ intersect. Set up and evaluate the double integral.

Computing Area with Green's Theorem

If we choose P and Q so that $Q_x - P_y = 1$, then $\oint P dx + Q dy = \text{Area}(D)$.

The cleanest choice: $P = -y/2$, $Q = x/2$, giving

$$\text{Area}(D) = \frac{1}{2} \oint_C x dy - y dx$$

This computes area from a boundary parameterization alone — no double integral.

Example 4: Area of an Ellipse

Example 4

Use the Green's Theorem area formula to find the area enclosed by the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

Parameterize the ellipse: $x = 3 \cos t$, $y = 2 \sin t$, $0 \leq t \leq 2\pi$. Compute dx and dy , then apply $A = \frac{1}{2} \oint x \, dy - y \, dx$.

Green's Theorem Explains Conservative Fields

Green's Theorem finally proves the “hard direction” of the cross-partials test from 16.3.

If $P_y = Q_x$ everywhere on a simply-connected region D , then for **any** closed curve C inside D :

$$\oint_C P \, dx + Q \, dy = \iint_D \underbrace{(Q_x - P_y)}_{=0} \, dA = 0$$

So $\vec{F}$ is path-independent, hence conservative.

Famous counterexample: $\vec{F} = \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$ has $P_y = Q_x$ everywhere except the origin, yet

$\oint \vec{F} \cdot d\vec{r} = 2\pi$ around any circle enclosing the origin. Green's Theorem doesn't apply because the hole breaks simple connectivity.

Simply connected + $P_y = Q_x \Rightarrow$ conservative. The hole in the domain is what allows $P_y = Q_x$ without conservative.

Homework

Section 16.4: 1, 3, 5, 9, 11, 15, 21